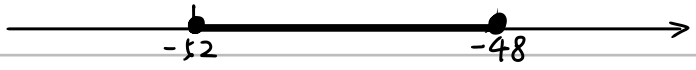
Calculus I Assignment 1

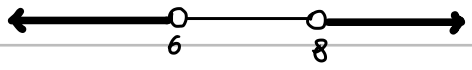
1. (a) $|x-3| \le 4 \iff -4 \le x-3 \le 4 \iff -1 \le x \le 7$



(b) $|x+50| \le 2 \iff -2 \le x+50 \le 2 \iff -52 \le x \le -48$



(c) $1 < |y-7| \iff y-7 > 1$ or $y-7 < -1 \iff y > 8$ or $y < 6$



(d) $|6-x| < 4 \iff -4 < 3-x < 4 \iff -7 < -x < 1$

$\iff -1 < x < 7$



2. (a) False. Counter-example: $a = 1/3$, $b = 1/2$

(b) False. $0 < a < b < 1 \Rightarrow 1/a > 1/b$

(c) True.

(d) False. Counter-example: $a = 1/4$, $b = 4/5$.

(e) True.

(f) False. Couter-example: $a = 1/\sqrt{2}$, $b = \frac{\sqrt{2}}{\sqrt{3}}$

(g) False Couter-example: $a = 1/\sqrt{3}$, $b = 1/\sqrt{2}$

(h) True. $0 < a < b < 1 \Rightarrow 1/a > 1 > b$

3. Since $\sqrt{3} \approx 1.732$, $|\sqrt{3} - 1.7| \approx 0.032$, we can choose $m \le 1$ such that $|\sqrt{3} - 1.7| < 10^{-m}$.

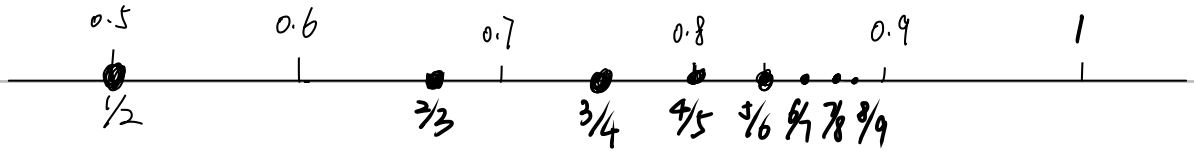
4.

	GLB	LUB
(a)	8	10
(b)	8	10
(c)	NOT existing	0
(d)	$\frac{30}{279}$	$\frac{59}{125}$
(e)	0	1

5. (a) Yes. An obvious upper bound is 1 (the area of the unit square).

(b) Yes. It is actually using the area of subsquares to approximate the area of the unit quarter circle. So the least upper bound of S is $\pi/4$.

6. (a)



(b) Let $\varepsilon > 0$. Because $\left|\frac{n}{n+1} - 1\right| = \frac{1}{n+1} < \varepsilon \iff n > \frac{1}{\varepsilon} - 1$, there exists $N = [\frac{1}{\varepsilon}]$ such that $\left|\frac{n}{n+1} - 1\right| \leq \left|\frac{N}{N+1} - 1\right| < \varepsilon$ when $n \geq N$. Hence $\lim_{n\to\infty} x_n = 1$.

(c)

ϵ	0.01	0.009	0.008	0.007	0.006	0.005	0.004	0.003	0.002	0.001
N	100	111	125	142	166	200	250	333	500	1000

7. (a) $0, \frac{2}{3}, 0, \frac{2}{3}, 0, \frac{2}{3}, 0, \frac{2}{3}, 0, \frac{2}{3}$ No limit.

(b) $-\frac{1}{2}, \frac{2}{3}, -\frac{3}{4}, \frac{4}{5}, -\frac{5}{6}, \frac{6}{7}, -\frac{7}{8}, \frac{8}{9}, -\frac{9}{10}, \frac{10}{11}$ No limit.

8 (a) Let $\varepsilon > 0$. Note that $\left|\frac{2n}{n^3+1} - 0\right| = \left|\frac{2n}{n^3+1}\right| < \frac{2}{n^2}$. Then because $\frac{2}{n^2} < \varepsilon \Leftrightarrow n > \sqrt{\frac{2}{\varepsilon}}$, there exists $N = [\sqrt{\frac{2}{\varepsilon}}] + 1$ such that $\left|\frac{2n}{n^3+1} - 0\right| < \frac{2}{n^2} \leq \frac{2}{N^2} < \varepsilon$ when $n \geq N$. Hence $\lim_{n\to\infty} \frac{2n}{n^3+1} = 0$.

(b) Let $\varepsilon > 0$. For $n \geq 2$, it holds that $\left|\frac{3n+5}{2n-3} - \frac{3}{2}\right| = \frac{19}{4n-6} < \varepsilon$ $\Leftrightarrow n > \frac{19+6\varepsilon}{4\varepsilon}$. So there exists $N = [\frac{19+6\varepsilon}{4\varepsilon}] + 1$ such that $\left|\frac{3n+5}{2n-3} - \frac{3}{2}\right| < \varepsilon$ when $n \geq N$. Hence $\lim_{n\to\infty} \frac{3n+5}{2n-3} = \frac{3}{2}$.

9. For $|q| < 1$, $|q|^n < |q|^N$ if $n > N$. Let $\varepsilon > 0$. Since $|q^n - 0| = |q|^n < \varepsilon \Leftrightarrow n \ln|q| < \ln\varepsilon \Leftrightarrow n > \frac{\ln\varepsilon}{\ln|q|}$, there exists $N = [\frac{\ln\varepsilon}{\ln|q|}] + 1$ such that $|q^n - 0| \leq |q^N - 0| < \varepsilon$ when $n \geq N$. Hence $\lim_{n\to\infty} q^n = 0$ for $|q| < 1$.

Bonus Questions

1. Let $\varepsilon > 0$. Without loss of generality, assume further that $0 < \varepsilon < 1$. Note that $|\sqrt[n]{a} - 1| < \varepsilon \Leftrightarrow -\varepsilon < \sqrt[n]{a} - 1 < \varepsilon \Leftrightarrow$ $1 - \varepsilon < \sqrt[n]{a} < 1 + \varepsilon \Leftrightarrow \ln(1-\varepsilon) < \frac{\ln a}{n} < \ln(1+\varepsilon) \Leftrightarrow n\ln(1-\varepsilon) < \ln a$ and $\ln a < n\ln(1+\varepsilon) \Leftrightarrow n > \frac{\ln a}{\ln(1-\varepsilon)}$ and $n > \frac{\ln a}{\ln(1+\varepsilon)}$. Then there

exists $N=\max\{[\frac{\ln a}{\ln(1+\varepsilon)}]+1, [\frac{\ln a}{\ln(1+\varepsilon)}]+1\}$ such that $|\sqrt[n]{a}-1|<\varepsilon$ when $n\geq N$. Hence $\lim_{n\to\infty}\sqrt[n]{a}=1$.

2. Let $\varepsilon>0$. Note that for $n\geq 2$, $|\frac{2^n}{n!}-0| = \frac{2}{(n-1)!}\leq\frac{2}{n-1}$. Since for $n\geq 2$, $\frac{2}{n-1}<\varepsilon \Leftrightarrow n>2/\varepsilon+1$, there exists $N=[2/\varepsilon]+2$ such that $|\frac{2^n}{n!}-0|\leq\frac{2}{n-1}\leq\frac{2}{N-1}<\varepsilon$ when $n\geq N$. Hence $\lim_{n\to\infty}\frac{2^n}{n!}=0$.

3. Let $\varepsilon>0$. Observe that $n^{1/n}>1$ and then

$$|n^{1/n}-1| = n^{1/n}-1<\varepsilon \Leftrightarrow n^{1/n}<\varepsilon+1 \Leftrightarrow n<(1+\varepsilon)^n$$
$$\Leftrightarrow n<1+n\varepsilon+\binom{n}{2}\cdot\varepsilon^2+\cdots+\binom{n}{n}\varepsilon^n \quad (*).$$

If $n<\binom{n}{2}\cdot\varepsilon^2=\frac{n(n-1)}{2}\cdot\varepsilon^2$, then $(*)$ holds. Note also that $n<\frac{n(n-1)}{2}\cdot\varepsilon^2 \Leftrightarrow \frac{2}{\varepsilon^2}<n-1 \Leftrightarrow n>2/\varepsilon^2+1$. We choose $N=[2/\varepsilon^2]+2$. Then when $n\geq N$, $n>2/\varepsilon^2+1$ and then $(*)$ holds, which implies $|n^{1/n}-1|<\varepsilon$.

Hence $\lim_{n\to\infty}\sqrt[n]{n}=1$.